

## # Force

In physics, a **force** is a push or pull that acts on an object as a result of its interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

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### ## Key Characteristics

- **Vector quantity** – Forces have both magnitude (strength) and direction.
- **Standard unit** – The metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of **1 m/s<sup>2</sup>**.

- **Formula** – Newton's Second Law of Motion:

[

$$F = m \cdot a$$

]

where

$(F)$  = force,

$(m)$  = mass,

$(a)$  = acceleration.

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### ## Common Types of Forces

- **Contact forces** – Act through physical contact between objects.

Examples: friction, tension, air resistance, push/pull.

- **Non-contact forces** – Act over a distance without physical contact.

Examples: gravity, electromagnetism, nuclear forces.

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### ### Quick Reference

| Property | Symbol | Unit | Example |
|----------|--------|------|---------|
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|       |       |   |                                                |
|-------|-------|---|------------------------------------------------|
| Force | $(F)$ | N | Weight of a 10 kg mass on Earth $\approx 98$ N |
|-------|-------|---|------------------------------------------------|

|      |       |    |      |
|------|-------|----|------|
| Mass | $(m)$ | kg | 5 kg |
|------|-------|----|------|

|              |       |                |                   |
|--------------|-------|----------------|-------------------|
| Acceleration | $(a)$ | $\text{m/s}^2$ | $2 \text{ m/s}^2$ |
|--------------|-------|----------------|-------------------|

Feel free to use these notes as a concise study guide for understanding the fundamentals of force in physics.